Cloning (digestion, dephosphorylation, ligation and concentration)

What you're going to get / What you expect

A recombinant pET plasmid (vector plus insert) concentrated and desalted, ready to transform into E. coli with a low self-ligation background.

Materials list (per reaction (insert 20 uL / vector as indicated))

- Purified insert (dsDNA) and pET vector.
- FastDigest NdeI and HindIII restriction enzymes, with FastDigest buffer.
- FastAP thermosensitive alkaline phosphatase.
- T4 DNA ligase and buffer.
- A gel and PCR clean-up kit (e.g. Wizard SV Gel and PCR Clean-Up System).
- SpeedVac centrifugal evaporator.
- Nitrocellulose microdialysis membranes (0.022 um pore).
- Nuclease-free water; 37 ºC and 16 ºC incubation.

Steps

1. Digest the inserts: incubate 1 ug of dsDNA with 1 uL of each restriction enzyme (NdeI and HindIII) in a final volume of 20 uL.
2. Digest the vector: use an initial mass of 2.5 ug of DNA under the same conditions.
3. Incubate both digestion reactions overnight at 37 ºC.
4. Dephosphorylate the digested vector immediately: add 1 uL of FastAP and incubate 30 minutes at 37 ºC, to stop the vector re-circularising on itself.
5. Purify all fragments with the gel and PCR clean-up kit.
6. Set the ligation with NEBioCalculator at a 1:5 vector:insert molar ratio; assemble a 20 uL reaction with T4 DNA ligase and incubate overnight at 16 ºC. Run a vector-only negative control in parallel.
7. Concentrate the ligation product in a SpeedVac for 10 minutes.
8. Desalt by microdialysis on a 0.022 um nitrocellulose membrane; use 2 uL of the dialysate to transform chemo-competent E. coli DH5a (see the transformation protocol).

Resources & Downloads

- NEBioCalculator: https://nebiocalculator.neb.com/
- Gel and PCR clean-up kit manual (e.g. Promega Wizard SV).

Troubleshooting

- Colonies on the vector-only control: confirm the FastAP dephosphorylation was done and that the vector was fully cut.
- Incomplete digestion: extend the incubation and verify both enzymes are active in the shared FastDigest buffer.
- If downstream identity checks fail despite bands of the expected size, prefer PCR-defined inserts over fragments excised from carrier plasmids.